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@PuckerLab

Prof. Dr. Boas Pucker

Long-Read Plant Genomics

@puckerlab



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Availability of slides

- All materials are freely available (CC BY) - after the lectures:
  - GitHub: https://github.com/bpucker/LRG

- Questions: Feel free to ask at any time

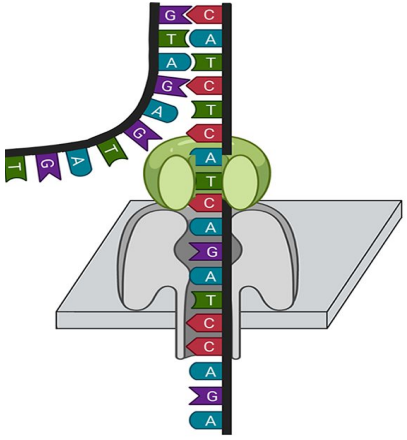
- Feedback, comments, or questions: pucker[a]uni-bonn.de

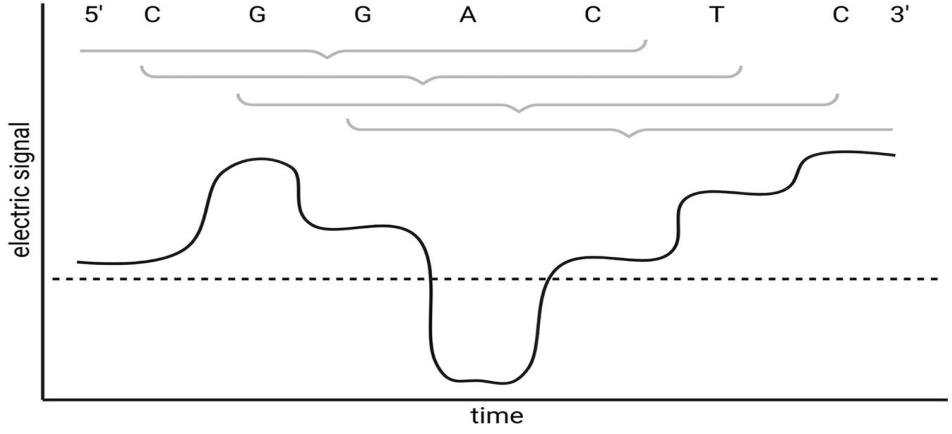
My figures and content can be re-used in accordance with CC BY 4.0, but this might not apply to all images/logos.

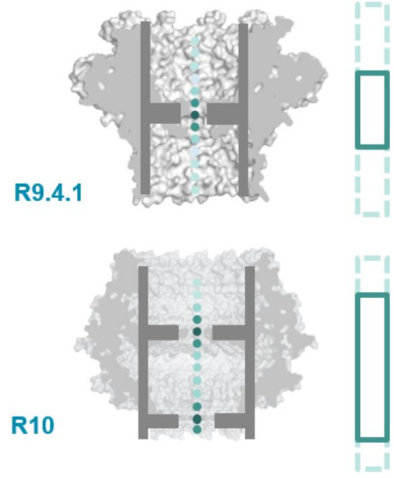
Oxford Nanopore Technologies (ONT)

Under development since the 1980s

“Analysis of anything, anywhere by anyone” … not restricted to DNA sequencing



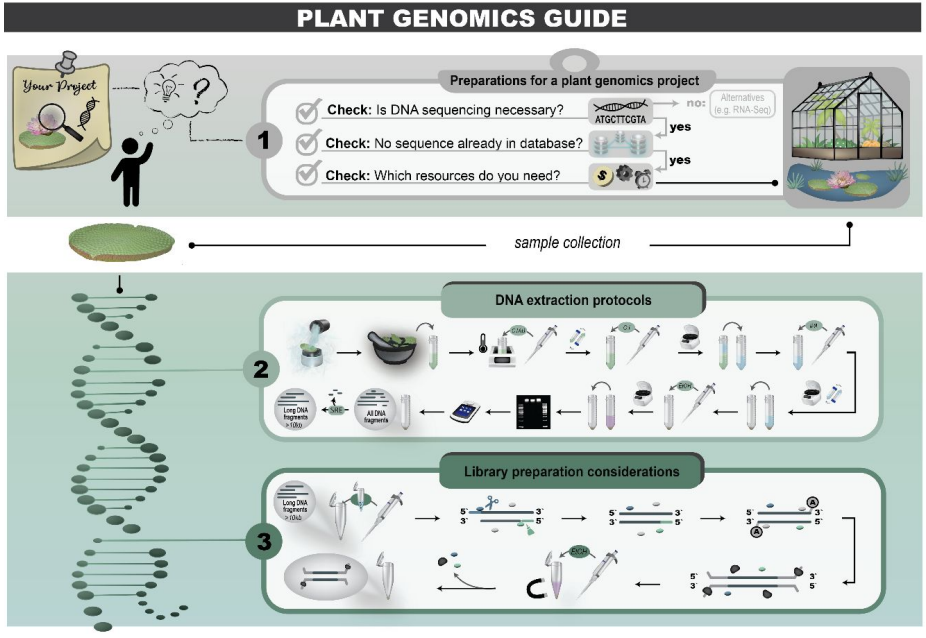


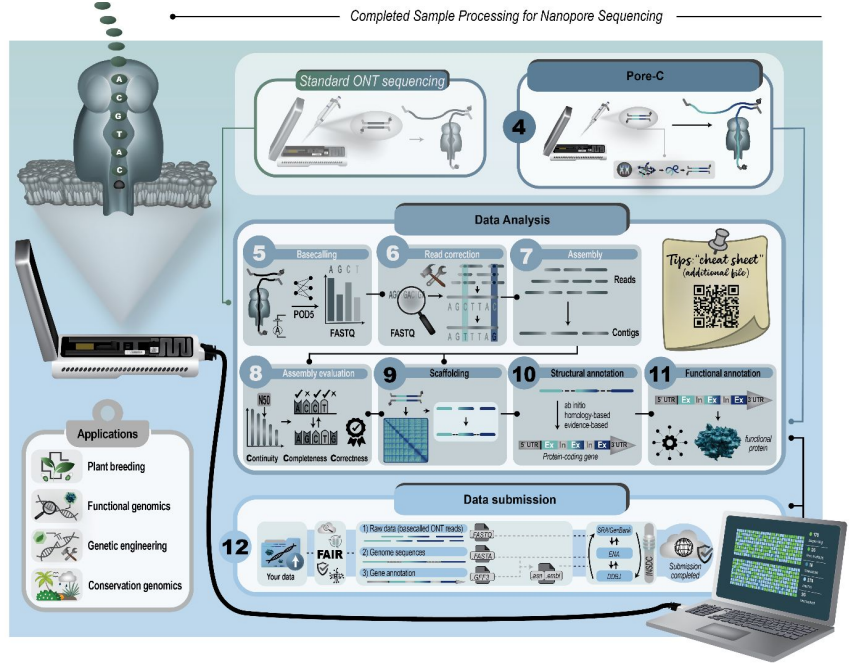


Pucker et al., 2022: 10.1017/qpb.2021.18
https://nanoporetech.com/about-us/news/r103-newest-nanopore-high-accuracy-nanopore-sequencing-now-available-store

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Established workflow for plant genomics



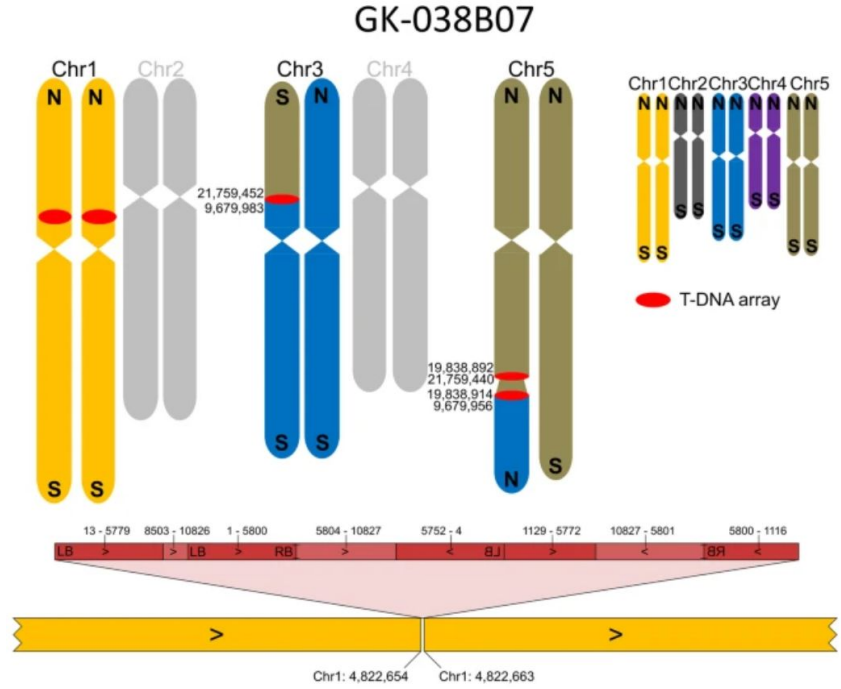


Supplementary file with all commands!

de Oliveira et al., 2026: 10.1186/s12864-026-12623-z

Long-read genotyping of T-DNA lines

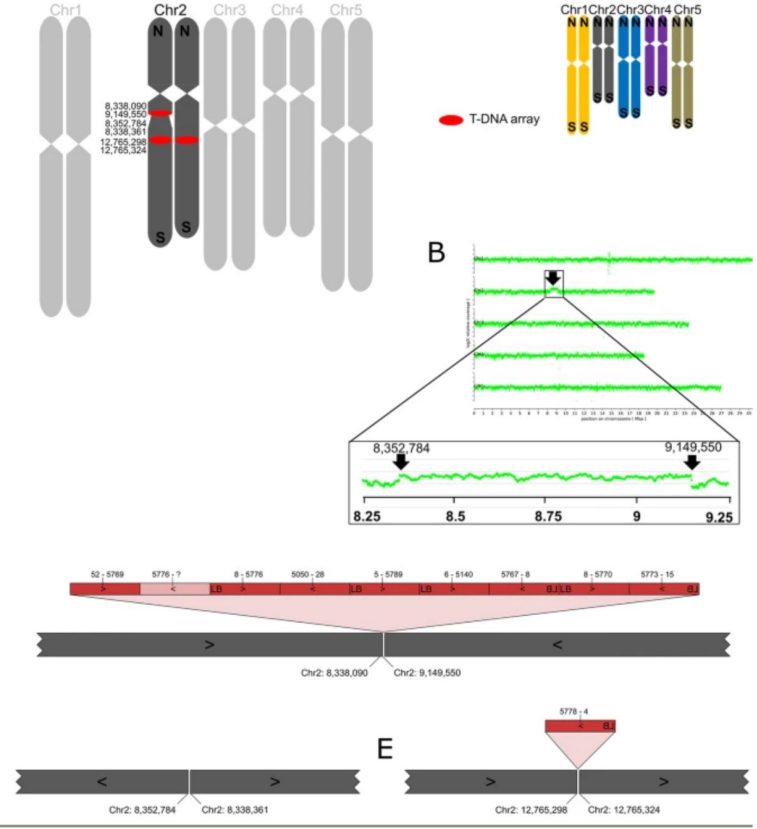
- T-DNA insertions generate knock-outs
- T-DNA insertion lines were backbone of gene function elucidation in A. thaliana
- GABI-Kat comprises >80,000 lines
- Genotyping with ONT sequencing reveals off-targets and number of T-DNAs



Pucker et al., 2021: 10.1186/s12864-021-07877-8

GK-767D12

- Photosynthesis is a fundamental property of plants
- Central genes of the photosynthesis cannot be knocked-out (would be lethal)
- GK-767D12 has a structural variant potentially affecting AT2G21385/CGLD11 (Chloroplast Grana-Less Development 11)



Pucker et al., 2021: 10.1186/s12864-021-07877-8

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Plan for today

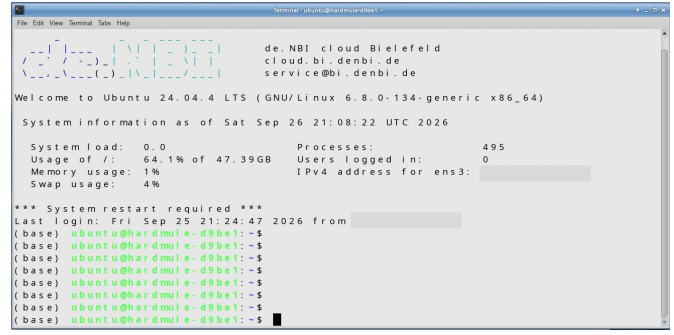
- Connect to a virtual machine (environment for computational analyses)
- Caclulate FASTQ file statistics
- Map reads against a reference genome sequence (TAIR10)
- Download the resulting BAM file
- Inspect your read mapping with IGV

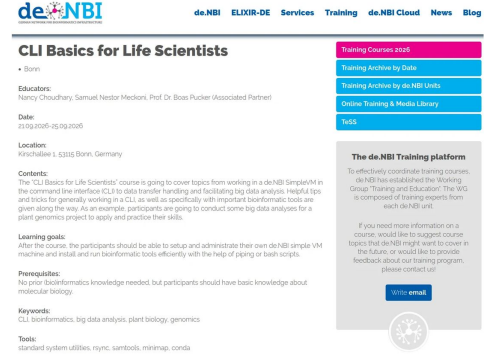
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de.NBI

GERMAN NETWORK FOR BIOINFORMATICS INFRASTRUCTURE

- Cloud computing resources for bioinformatics
- Hosting of services (e.g. pbb-tools.de)
- Training courses for a diversity of methods
- Network for exchange about latest developments in computational biology





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Storing sequencing data in FASTQ

- Standard format for sequencing data
- Four lines per entry:
  - Header starts with @ (title + description)
  - Sequence
  - + (optional repetition of header)
  - Quality (phred encoded in ASCII character)
- Different versions existed (outdated today)
- Example:

@seq1

ACGTACGTACGT

+

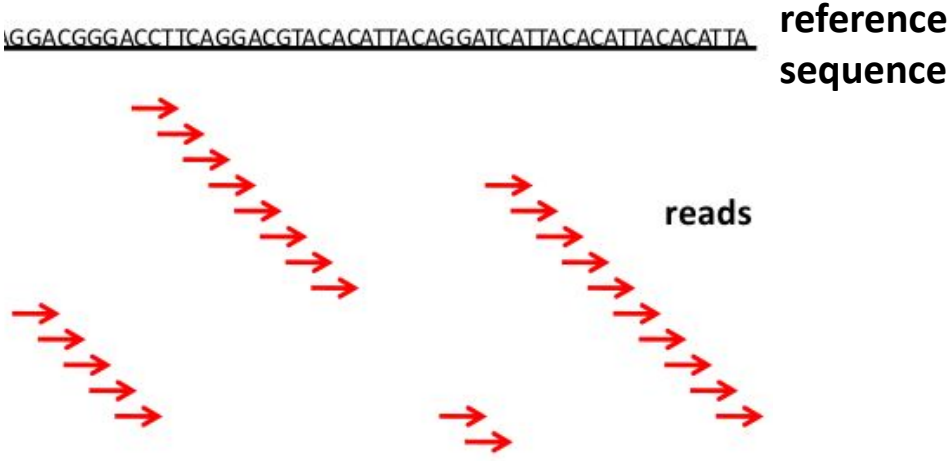
“”?CB””:DC”

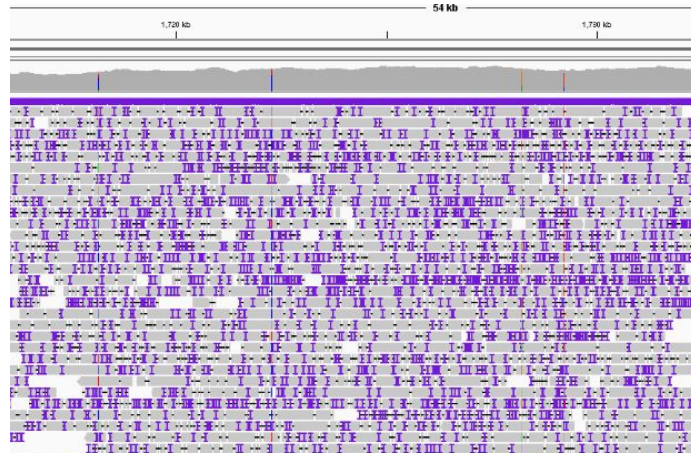
Symbol	Phred Quality Score	Probability of Incorrect Base Call
!	0	1.000
"	1	0.794
#	2	0.631
$	3	0.501
%	4	0.398
&	5	0.316

Phred quality score	Error probability	Accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1,000	99.9%
40	1 in 10,000	99.99%
50	1 in 100,000	99.999%

Read mapping: concept & reality

- Alignment of reads against a reference sequence
- SAM = Sequence Alignment/Map
- BAM = Binary Alignment Map





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Integrative Genomics Viewer (IGV)

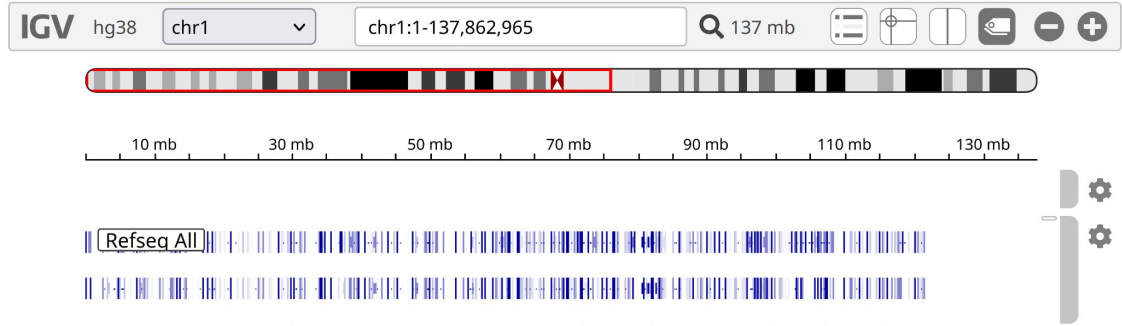
- Download IGV from:

  https://igv.org/doc/desktop/

- or use IGV online:

  https://igv.org/app/

- Load genome from file
- Load file (select BAM file)
- Load file (select GFF file)



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Starting VM

- Obtain a private key file (matching to public key file in VM)
- Download the README file
- Follow instructions in the README file



https://uni-bonn.sciebo.de/s/SENw6ZYqZgsjFa2

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Exercises

Please find the commands for all exercises here:

https://github.com/bpucker/LRG/blob/main/README_EXERCISES1.md



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Backup: obtaining results

- Download:
  - FASTA + FAI file
  - GFF file
  - BAM + BAI file

- Visualize read mapping in IGV



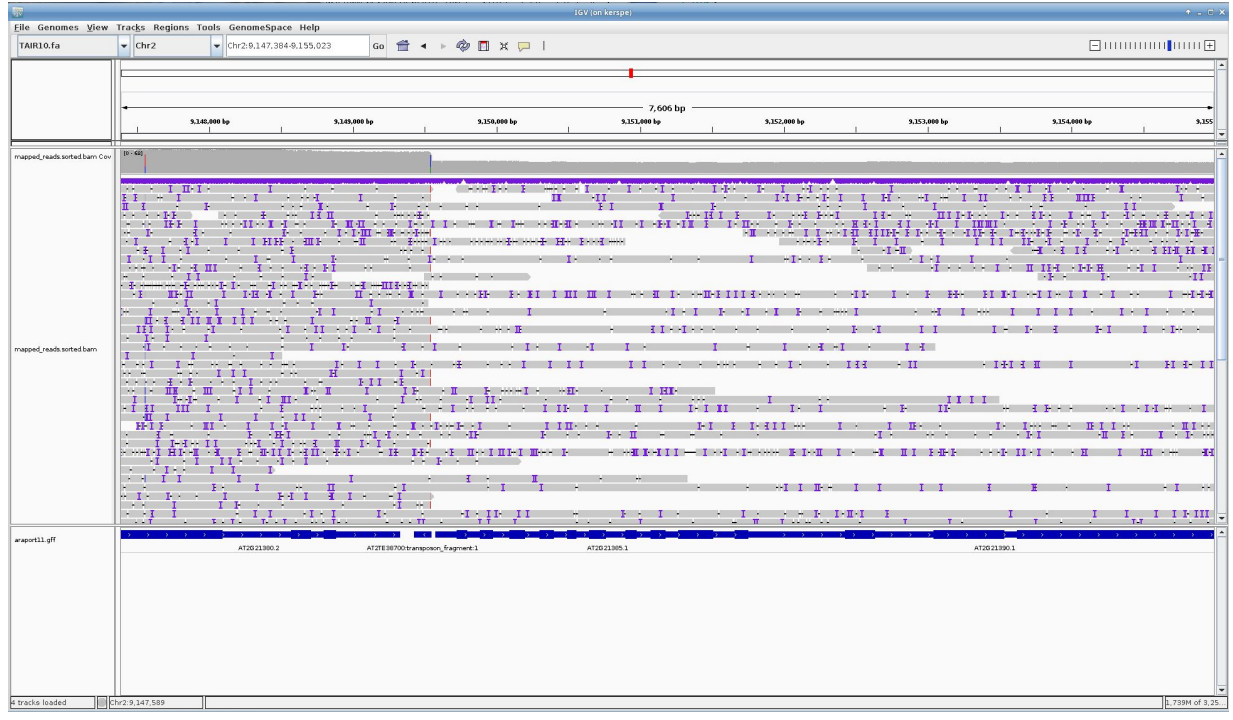
https://uni-bonn.sciebo.de/s/YNoKe4b3XGbwMQ4

Inspecting read mapping results

GK-767D12 = ERR3712254

AT2G21385:

Chr2 9149577 9151970



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Time for questions!